

Skills Summary

One-Variable Inequalities: Reversing the Symbol

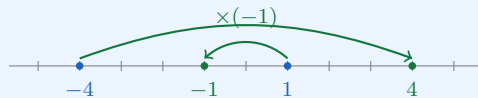
Use this reference while completing your worksheet or when studying.

1. Multiply or divide by a negative

The rule: multiplying or dividing *both sides* by a **negative** number **reverses** the symbol. Adding or subtracting anything — including a negative — does *not*.

$>$ becomes $<$ $<$ becomes $>$ $\geq$ becomes $\leq$ $\leq$ becomes $\geq$

Why: multiplying by a negative *reflects* the number line through 0. Whatever was on the left lands on the right, so “smaller” and “larger” trade places.



Before: $-4 < 1$. After the reflection: $4 > -1$. Same two numbers, opposite order.

Example: Solve $-6x < 18$

Step 1. The variable is multiplied by -6 , so undoing it means dividing by a negative — that is the one move that reverses the symbol.

Step 2. $\frac{-6x}{-6} \overset{\text{flip}}{>} \frac{18}{-6} \Rightarrow x > -3$

Step 3. Check with a number from the set: $x = 0$ gives $-6(0) = 0 < 18 \checkmark$.

Try it: Solve $-2x \geq 10$

check: $-5 > x$

2. Isolate first, then flip

The order: (1) add or subtract to get the x -term alone — this step *never* flips; (2) divide by the coefficient — this step flips *only if* that coefficient is negative.

A negative sign inside the problem is not enough. Ask the narrower question: “**am I dividing by a negative right now?**”

Example: Solve $5 - 4x \leq 17$

Step 1. Two moves are needed and only the second one can flip, so subtract first and decide about the symbol later.

Step 2. Subtract 5: $-4x \leq 12$ (symbol unchanged — this was a shift).

Step 3. Divide by -4 : $\frac{-4x}{-4} \overset{\text{flip}}{\geq} \frac{12}{-4} \Rightarrow x \geq -3$

Try it: Solve $2 - 5x > 22$

check: $-4 > x$

3. Diagnose a missing flip

The two-second test: pick any number from the answer's solution set and put it into the **original** inequality.

- If it makes a true statement, the answer is probably right.
- If it makes a *false* statement, and the arithmetic checks out, the symbol is backwards — a missing flip.

Test a number *far* from the boundary (-10 , not -2.9): a wrong symbol shows up loudest out there.

Example: Ana solved $-8x > 24$ and wrote $x > -3$. Diagnose and fix.

Step 1. Her arithmetic ($24 \div -8 = -3$) is right, so test her solution set rather than re-doing the division.

Step 2. Take $x = 0$ from her set: $-8(0) = 0$, and $0 > 24$ is false — her set holds non-solutions, so the symbol is backwards.

Step 3. Dividing by -8 reverses it: $x < -3$

Try it: Rob solved $-\frac{x}{2} \leq 4$ and wrote $x \leq -8$. Fix it.

check: $-\frac{-8}{2} \leq 4$

(!) Watch out: A negative sign somewhere in the problem does not by itself flip anything. In $x - 7 > -2$ you add 7 to both sides and the symbol stays: $x > 5$. The flip is caused only by *multiplying or dividing* both sides by a negative.